



$$P(\text{真}) = \frac{1}{2}P + \frac{3}{10}(1-P) = \frac{3}{10} + \frac{2}{10}P$$

$$P(\text{假}) = \frac{3}{10}P + \frac{1}{2}(1-P) = \frac{1}{2} - \frac{1}{10}P$$

$$P(\text{中}) = \frac{2}{10}P + \frac{1}{10}(1-P) = \frac{1}{10}$$

设 X, Y 两随机变量分别代表某人村属、说话真假.

由题, 应求的是 X, Y 的互信息, 即 $I(X; Y)$

$$I(X; Y) = H(Y) - H(Y|X) \quad H(Y|X) = P \cdot H(Y|A) + (1-P) \cdot H(Y|B)$$

由于 A, B 关于真假的分布在数值上一致, $\therefore H(Y|A) = H(Y|B) \quad \therefore H(Y|X) = H(Y|B) = H(Y|A)$

$$H(Y) = -\sum P(y) \log_2 P(y) = -\left[\left(\frac{3}{10} + \frac{2}{10}P \right) \log_2 \left(\frac{3}{10} + \frac{2}{10}P \right) + \left(\frac{1}{2} - \frac{2}{10}P \right) \log_2 \left(\frac{1}{2} - \frac{2}{10}P \right) + \frac{2}{10} \log_2 \frac{2}{10} \right]$$

$$\therefore I(X; Y) = -(0.3 + 0.2P) \log_2 (0.3 + 0.2P) + (0.2P - 0.5) \log_2 (0.5 - 0.2P) - 0.2 \log_2 0.2 + 0.5 \log_2 0.5 + 0.3 \log_2 0.3 + 0.2 \log_2 0.2$$

当 $P(\text{真}) = P(\text{假})$ 时, 熵最大. $0.3 + 0.2P = 0.5 - 0.2P \Rightarrow P = 0.5$

$$\therefore I_{\max}(X; Y) = -1.6 + 0.3 \log_2 3 + 0.5 \log_2 0.5$$

2.2. 设随机变量 X, Y 分别代表骰子点数和硬币正面次数

$$I(X; Y) = H(Y) - H(Y|X)$$

$$H(Y|X) = -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) \frac{2}{3} - \left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} \right) \frac{1}{3} = \frac{7}{6} \text{ bits}$$

$$H(Y) = -\left(\frac{5}{12} \log_2 \frac{5}{12} + \frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{12} \log_2 \frac{1}{12} \right) = 1.5 + 0.5 \log_2 3 - \frac{5}{12} \log_2 5 \text{ bits}$$

$$\therefore I(X; Y) = \frac{1}{3} + \frac{1}{2} \log_2 3 - \frac{5}{12} \log_2 5 \text{ bits}$$

$$\begin{cases} P(Y=0|X) = \frac{1}{2} & X \in \{1, 2, 3, 4\} \\ P(Y=0|X) = \frac{1}{4} & X \in \{5, 6\} \\ P(Y=1|X) = \frac{1}{2} & X \in \{1, 2, 3, 4\} \\ P(Y=1|X) = \frac{1}{2} & X \in \{5, 6\} \\ P(Y=2|X) = 0 & X \in \{1, 2, 3, 4\} \\ P(Y=2|X) = \frac{1}{4} & X \in \{5, 6\} \end{cases}$$

2.4. 将三颗骰子点数分别视为独立的三个随机变量 D_1, D_2, D_3 . 它们互相独立. $P(D_i) = \frac{1}{6}$.

$$X = D_1 \quad Y = X + D_2 \quad Z = Y + D_3$$

$$H(Y|X) = H(X + D_2|X) = H(D_2|X) = H(D_2) = -6 \cdot \frac{1}{6} \log_2 \frac{1}{6} = \log_2 6 \text{ bits}$$

$$\text{同理 } H(Z|X, Y) = H(Y + D_3|X, Y) = H(D_3|X, Y) = H(D_3) = \log_2 6 \text{ bits}$$

$$H(X|Y) = \sum_{y=2}^{12} P(y) \cdot H(X|Y=y) = \sum_{y=2}^{12} \frac{K_y}{36} \log_2 K_y = \frac{2}{36} (\log_2 1 + 2 \log_2 2 + 3 \log_2 3 + 4 \log_2 4 + 5 \log_2 5) + \frac{6}{36} \log_2 6$$

$$= \frac{13}{18} + \frac{1}{3} \log_2 3 + \frac{5}{18} \log_2 5 \text{ bit}$$

$$H(X, Z|Y) = H(X|Y) + H(Z|X, Y) = \frac{31}{18} + \frac{4}{3} \log_2 3 + \frac{5}{18} \log_2 5 \text{ bit}$$

$$H(Z|X) = H(X + D_2 + D_3|X) = H(D_2 + D_3) = H(Y)$$

$$\therefore H(X, Y) = H(D_1, D_2) = H(D_1) + H(D_2) = 2 \log_2 6 = H(Y) + H(X|Y)$$

$$\therefore H(Y) = 2 \log_2 6 - H(X|Y) = \frac{23}{18} + \frac{5}{3} \log_2 3 - \frac{5}{18} \log_2 5 \text{ bit}$$

2.5 设 x, y 分别代表发送的数字和接收的数字.

当 x 为偶数时 $P(Y=x|X=x) = 1$

当 x 为奇数时 $P(Y=x|X=x) = 0.5$ $P(Y=y|X=x) = \frac{0.5}{4} = 0.125$ (y 为与 x 不同的其他奇数)

$$H(Y|X) = \sum_{x=0}^9 P(X=x) \cdot H(Y|X=x) = \frac{1}{10} \sum_{x=0}^9 H(Y|X=x) = \frac{1}{10} \sum_{x=0}^9 H(Y|X=x) = \frac{1}{2} H(Y|X=1) = -\frac{1}{2} \sum_{y=0}^9 P(Y=y) \log_2 P(Y=y)$$

$$= -\frac{1}{2} \left(\frac{1}{2} \log_2 \frac{1}{2} + 4 \cdot \frac{1}{8} \log_2 \frac{1}{8} \right) = 1 \text{ bit.}$$

$$P(Y=y) = \frac{1}{10} \quad (\text{当 } y \text{ 为偶数时})$$

由对称性得 $P(Y=y) = \frac{1}{10}$ (当 y 为奇数时).

$$\therefore H(Y) = \log_2 10 \text{ bits}$$

$$\therefore I(X;Y) = H(Y) - H(Y|X) = \log_2 5 \text{ bits}$$

2.6.

$$H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$$

$$H(X,Y|Z) = H(Y|Z) + H(X|Y,Z)$$

$$\therefore H(X|Y,Z) \leq H(X|Y)$$

$$\therefore H(X,Y|Z) \leq H(Y|Z) + H(X|Y)$$

下证 $H(X,Y|Z) \geq H(X|Z)$ 显然, 故 (a) 得证.

对换示绪出的不等式右边: 设 $t = H(X|Z) + H(X|Y)$

$$RHS = \frac{t}{t+H(Z)} = 1 - \frac{H(Z)}{t+H(Z)} \text{ 并递增.}$$

$$\text{由 (a) } t \geq H(X|Z) \quad \therefore RHS \geq 1 - \frac{H(Z)}{H(X|Z)+H(Z)} = 1 - \frac{H(Z)}{H(X,Z)}$$

$$\therefore LHS \geq 1 - \frac{H(Z)}{H(X,Z)} \quad \text{下证 } 1 - \frac{H(Z)}{H(X,Z)} \geq \frac{H(X|Z)}{H(X,Z)}$$

$$\text{即证 } H(X,Z) - H(Z) \geq H(X|Z)$$

$$\Rightarrow H(X,Z) \geq H(Z) + H(X|Z) \text{ 显然成立.}$$

故 (b) 得证.

2.9.

$$(a) \quad H(Z) \geq H(X+Y|Y) = H(X|Y) = H(X)$$

$$(b) \quad \text{同理 } H(Z) \geq H(Y)$$

$$(c) \quad \text{要证 } H(X,Y) \geq H(Z) \text{ 即证 } H(X) + H(Y) \geq H(X+Y) = H(X+Y, X) - H(X|X+Y)$$

$$\text{又 } H(X+Y, X) = H(Y, X) = H(X) + H(Y)$$

即证 $H(X|X+Y) \geq 0$ 显然成立, 故得证.

$$(d) \quad \text{要证 } I(X;Z) = H(X) - H(X|Z) = H(Z) - H(Y)$$

$$\text{只要证 } H(X) + H(Y) = H(X|Z) + H(Z)$$

即 $H(X,Y) = H(X,Z) = H(X, X+Y)$ 显然成立 (因为对每组 (X,Y) 都对应唯一 $(X, X+Y)$).

要证

$$(e) I(X, Y; Z) = H(Z) - H(Z|X, Y)$$

∵ 已知了 X, Y 后 $Z = X + Y$ 确定了. $\therefore H(Z|X, Y) = 0$

$$\therefore I(X, Y; Z) = H(Z) \quad \text{得证}$$

$$I(X, Y; Z) = H(X, Y) - H(X, Y|Z) = H(X) + H(Y) - (H(X|Z) + H(Y|X, Z))$$

$H(X) \neq H(X|Z)$: 不独立!

不要以为 Z, X 独立 $\text{Cov}(X, Z) = \text{Var}(X) \neq 0$

故 X, Z 可能不独立!

(f)

$$I(X; Y, Z) = H(X) - H(X|Y, Z)$$

下证 $H(X|Y, Z) = 0$.

$$H(X|Y, Z) + H(Y, Z) = H(X, Y, Z) = H(X, Y) + \underbrace{H(Z|X, Y)}_{=0}$$

$$\therefore H(X|Y, Z) + H(Y, Z) = H(X, Y)$$

$$\therefore H(Y, Z) = H(X, Y)$$

$$\therefore H(X|Y, Z) = 0$$

$$\therefore I(X; Y, Z) = H(X)$$

$$(g) I(Y; Z|X) = H(Y|X) + H(Z|X, Y) = H(Y|X) = H(Y) \quad \text{得证}$$

$$(h) I(X, Y|Z) = H(X|Z) + H(Y|X, Z) \quad \text{由 f 得 } H(X|Y, Z) = 0 \text{ 同理 } H(Y|X, Z) = 0$$

$$\therefore I(X, Y|Z) = H(X|Z)$$

$$\text{由对称性 } I(X, Y|Z) = H(Y|Z)$$

2.12

$$(a) H(Y, Z|X) = H(Y|X) + H(Z|X, Y) = H(Y|X) + H(Z|X)$$

$$\text{即 } H(Z|X) = H(Z|X, Y)$$

$$H(Z|X) = \sum_x \sum_z p(x, z) \log_2 p(z|x) = \sum_x \sum_y \sum_z p(x, y, z) \log_2 p(z|x)$$

$$H(Z|X, Y) = \sum_x \sum_y \sum_z p(x, y, z) \log_2 p(z|x, y)$$

$$\therefore \sum_x \sum_y \sum_z p(x, y, z) \log_2 \frac{p(z|x)}{p(z|x, y)} = 0$$

$$\log_2 \frac{p(z|x)}{p(z|x, y)} = \log_2 \frac{p(z|x) p(x, y)}{p(x, y, z)} = \log_2 \frac{\frac{p(x, z)}{p(x)} \cdot \frac{p(x, y)}{p(x)}}{\frac{p(x, y, z)}{p(x)}} = \log_2 \frac{p(z|x) \cdot p(y|x)}{p(y, z|x)}$$

$$\sum_x \sum_y \sum_z p(x, y, z) \log_2 \frac{p(z|x) \cdot p(y|x)}{p(y, z|x)} = \sum_x p(x) \cdot \sum_y \sum_z p(y, z|x) \log_2 \frac{p(y, z|x)}{p(z|x) \cdot p(y|x)}$$

$$D_{KL}(p||q) \geq 0$$

符合KL散度的公式

当 $p(y, z|x) = p(z|x) \cdot p(y|x)$ 时成立

取 "="

$$\therefore \text{对 } \forall i, j, k. p(y_j, z_k|x_i) = p(y_j|x_i) p(z_k|x_i) \text{ 成立}$$

下证必要性: $\therefore \text{对 } \forall i, j, k. p(y_j, z_k|x_i) = p(y_j|x_i) p(z_k|x_i) \text{ 成立. 即 } p(y, z|x) = p(y|x) \cdot p(z|x) \text{ 成立.}$

$$H(Y, Z|X) = -\sum_{x,y,z} p(x, y, z) \log_2 p(y, z|x) = -\sum_{x,y,z} p(x) \cdot p(y, z|x) \log_2 [p(y|x) \cdot p(z|x)]$$

$$= \sum_{x,y,z} (-p(x) p(y, z|x) \log_2 p(y|x) - p(x) \cdot p(y, z|x) \log_2 p(z|x))$$

$$= \sum_{xy} p(x) p(y|x) p(z|x) \log_2 p(y|x) - \sum_{xy} p(x) \cdot p(y|x) \cdot p(z|x) \log_2 p(z|x)$$

$$= - \sum_{xy} p(x, y) \log_2 p(y|x) - \sum_{xy} p(x, y) \log_2 p(z|x)$$

$$= H(Y|x) + H(Z|x) \quad \text{得证.}$$

$$(b) H(Y, Z|x) = \sum_{xyz} p(x, y, z) \log_2 p(y, z|x)$$

$$H(Y|x) = \sum_{xy} p(x, y) \log_2 p(y|x) = \sum_{xyz} p(x, y, z) \log_2 p(y|x)$$

$$H(Z|x, Y) = \sum_{xyz} p(x, y, z) \log_2 p(z|x, y)$$

$$H(Y|x) + H(Z|x, Y) = \sum_{xyz} p(x, y, z) \log_2 [p(z|x, y) \cdot p(y|x)] = \sum_{xyz} p(x, y, z) \log_2 \frac{p(x, y, z)}{p(x)} \\ = H(Y, Z|x) \quad \text{得证.}$$

(C) 充分性:

$$H(Z|X, Y) = H(Z|Y)$$

$$\therefore \sum_{xyz} p(x, y, z) \log_2 p(z|y, x) = \sum_{xyz} p(x, y, z) \log_2 p(z|y)$$

$$\sum_{xyz} p(x, y, z) \log_2 \frac{p(x, y, z) p(y)}{p(y, z) \cdot p(x, y)} = \sum_{xyz} p(x, y, z) \log_2 \frac{p(x, z|y)}{p(x|y) p(z|y)} = \sum_y p(y) \sum_{xz} p(x, z|y) \log_2 \frac{p(x, z|y)}{p(x|y) p(z|y)} = 0$$

KL 散度

仅在 $p(x, z|y) = p(x|y) p(z|y)$ 处处相等时取 "=".

故充分性得证.

必要性:

$$\text{已知 } p(x, z|y) = p(x|y) \cdot p(z|y)$$

$$H(Z|X, Y) - H(Z|Y) = \sum_{xyz} p(x, y, z) \log_2 \frac{p(z|x, y)}{p(z|y)} = \sum_{xyz} p(x, y, z) \log_2 1 = 0$$

$$\therefore H(Z|X, Y) = H(Z|Y) \quad \text{成立.}$$

综上, 命题得证.

2.13

$$(a) H(x) = - \sum_{i=1}^m p(x=i) \log_2 p(x=i) - \sum_{j=1}^n p(x=j) \log_2 p(x=j)$$

$$\text{对于 } 1-m \quad p(x=x) = \alpha p_1(x)$$

$$\text{对于 } m+1-n \quad p(x=x) = (1-\alpha) p_2(x)$$

$$\therefore H(x) = - \sum \alpha p_1(x) \log_2 (\alpha p_1(x)) - \sum (1-\alpha) p_2(x) \log_2 [(1-\alpha) p_2(x)]$$

$$= -\alpha \log_2 \alpha \sum p_1(x) - \alpha \sum p_1(x) \log_2 p_1(x) - (1-\alpha) \log_2 (1-\alpha) \sum p_2(x) - (1-\alpha) \sum p_2(x) \log_2 p_2(x)$$

$$= H(\alpha) + \alpha H(x_1) + (1-\alpha) H(x_2)$$

$$(b) \frac{dH(x)}{d\alpha} = \log_2 \frac{1-\alpha}{\alpha} + H(x_1) - H(x_2) = 0 \quad \text{得 } \alpha^* = \frac{2^{H(x_1)}}{2^{H(x_1)} + 2^{H(x_2)}} \quad \text{且 } A = 2^{H(x_1)} \quad B = 2^{H(x_2)}$$

$$\text{将 } \alpha^* \text{ 代入 } H(x) \text{ 得 } H(x)_{\max} = \frac{-A}{A+B} \log_2 \frac{A}{A+B} - \frac{B}{A+B} \log_2 \frac{B}{A+B} + \frac{A}{A+B} \log_2 A + \frac{B}{A+B} \log_2 B = \log_2 (A+B) = \log_2 (2^{H(x_1)} + 2^{H(x_2)})$$

$$\therefore 2^{H(x)_{\max}} \leq 2^{H(x_1)} + 2^{H(x_2)}$$

证毕!

